27/04

Storage classes:

In C we have 4 storage classes

1. auto — gar — with in the blocks — comes with fun, end with fun — stacks
2. register — gar — " — " — " — regvar
3. static
   - static local — 0 — with in the fun — as long as prg in exe — DS
   - static global — 0 — with in the file — as long as prg in exe — DS
4. extern — 0 — with in the prg — as long as prg in exe — DS

In C++ we have 5 storage classes

1. auto
2. register
3. static
4. extern

} Same as C

5. mutable ----> mutable storage class is applicable only to the members of class.

Structure :

```
void foo()
{ Struct Bank W;

Struct Bank
{
private: int accno;
         char name[20];
         float bal;
public:  void Deposit(float amt)
         {
           bal += amt;
         }
         bool Withdrawal(float amt)
         { if(amt <= bal)
           { bal -= amt; return true; }
           else { cout << "insufficient funds"; return false; }
         }
};
```

Encapsulation: Binding of data and its functions into a single entity is called Encapsulation.

Abstraction : Hidding the implementation details and giving only essential details is called Abstraction.

In C++ encapsultion and abstraction are implemented using class and structures

What is access specifiers?
Access specifiers decide who can access the members of a class or structure.
We have 3 access specifiers
1. private
2. protected
3. public

private : private members of aclass or structure can be accessible by members of a class or friends of a class.
protected : protected members of a class are accessible by members of a class , friends of a class and child classes.
public : members of a class as well as non members of a class can access

What is a class?
A class is a userdefined datatype same linke structures, Which is collection of variables and functions.
Aclass is a datatype or class is abstract datatype.

A class is ablue print of an object.
A class is just a logical existence

What is an object ?
An object is an instant of a class.
An object is a physical entity of a class.

class	object
fruit	mango,apple,grapes
Color	white,pink,blue,yellow,black
v23he7	v23he7A1,m1,a4,k4,

syntax of class

```
class classname
{
        access specifiers: members of class
};
```

ex :

```
class ABC
{
        int data1,data2;
        public : void f1(){ int a ----}
                 void f2(){ int b-----}
};
```

class declaration means designing a new data type called ABC no memory is allocated or consumed.

syntax of object creation:

```
classname objectname;
```

ex : ABC obj;

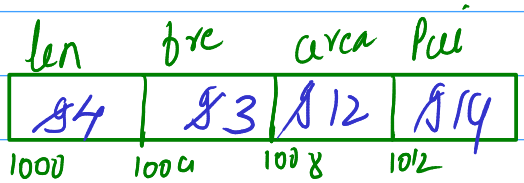
ABC → datatype
obj → object

sizeof an object
sum of the sizes of the datamembers of a class.

Variables of a class are called data members
Functions inside the class or functions of a class are called member functions.

Both together are called Members of a class.

refer to Rectangle.cpp




```
Rectangle robj;
robj.SetData();
robj.CalArea();
robj.CalPeri();
```

data membersv inside the member functions belows to calling object.
What is calling object?
An object invoking the function is called calling object.
ex:

```
robj.SetData();
```

↳ calling object

```
sobj.Print();
```

↳ calling object